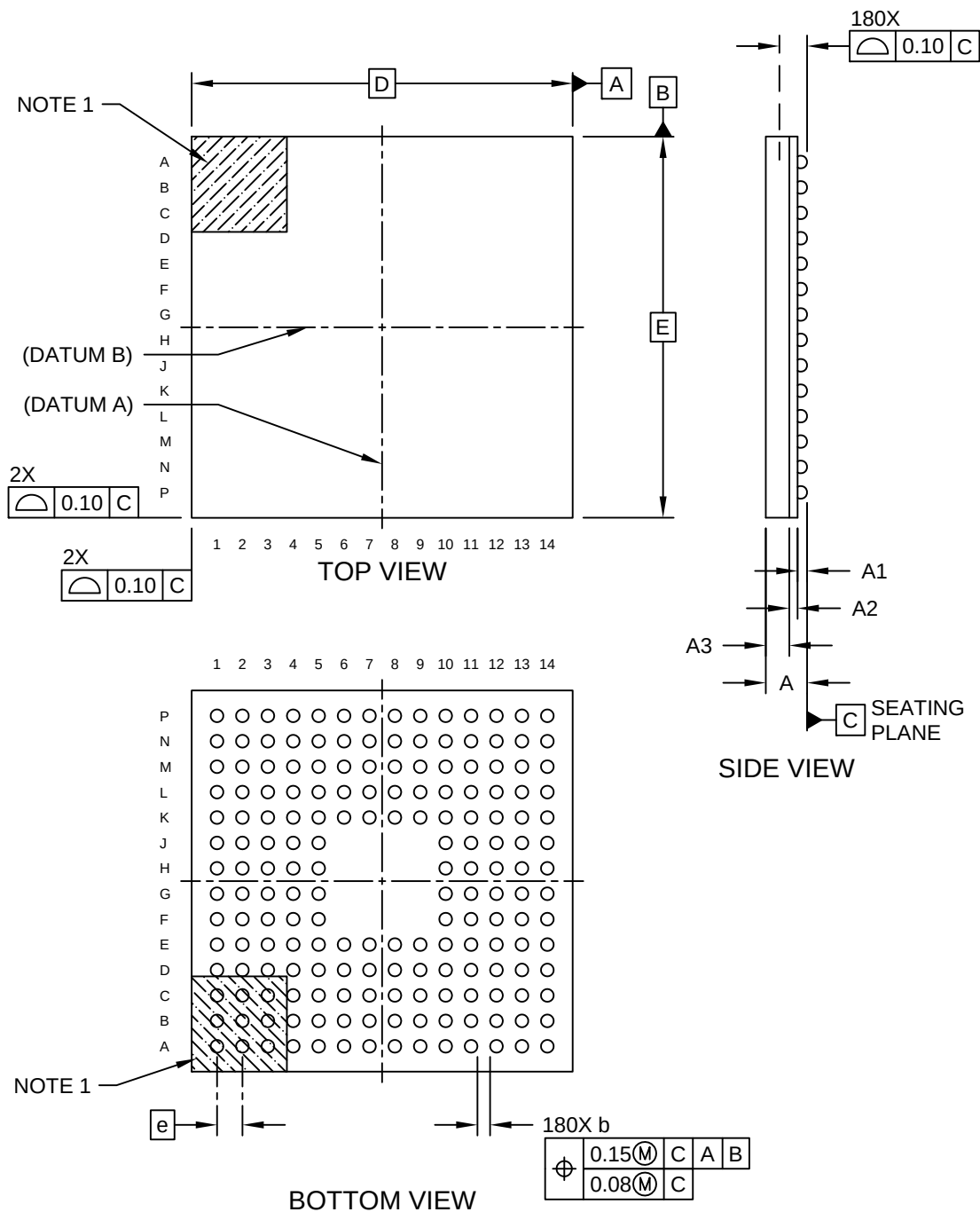


180-Ball Low Profile Fine Pitch Ball Grid Array (AGB) - 12x12x1.3mm Body [LFBGA] Atmel Legacy Global Package Code CDZ

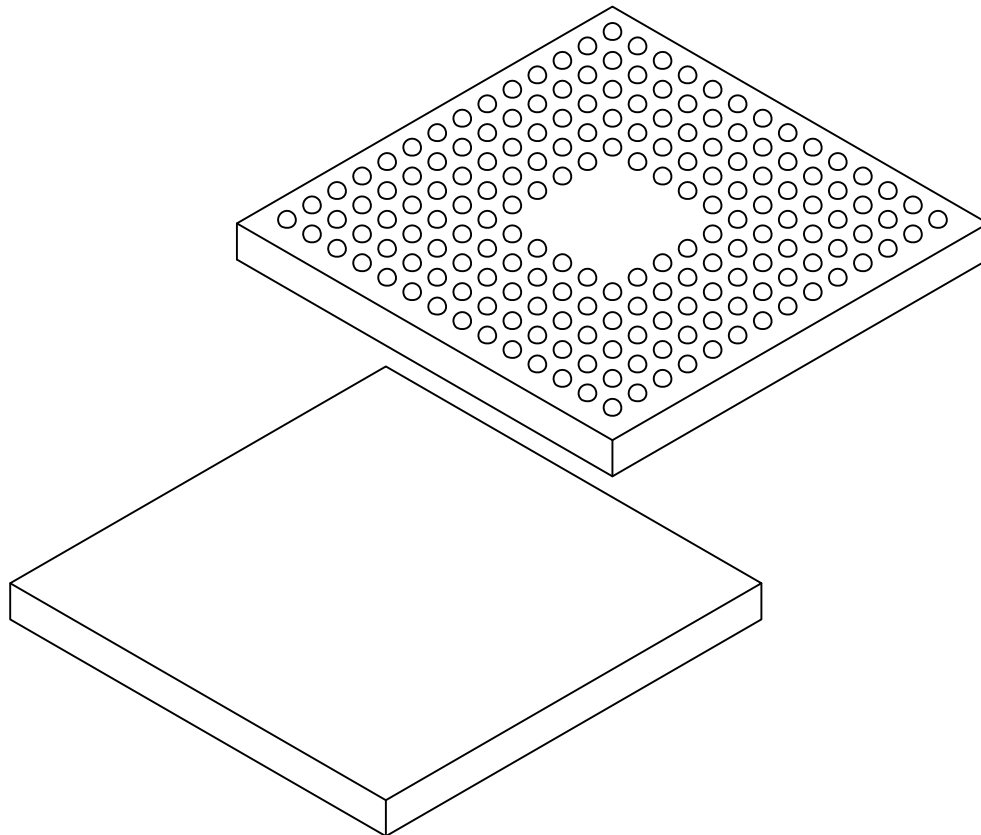
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-21120-AGB Rev A Sheet 1 of 2

180-Ball Low Profile Fine Pitch Ball Grid Array (AGB) - 12x12x1.3mm Body [LFBGA] Atmel Legacy Global Package Code CDZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



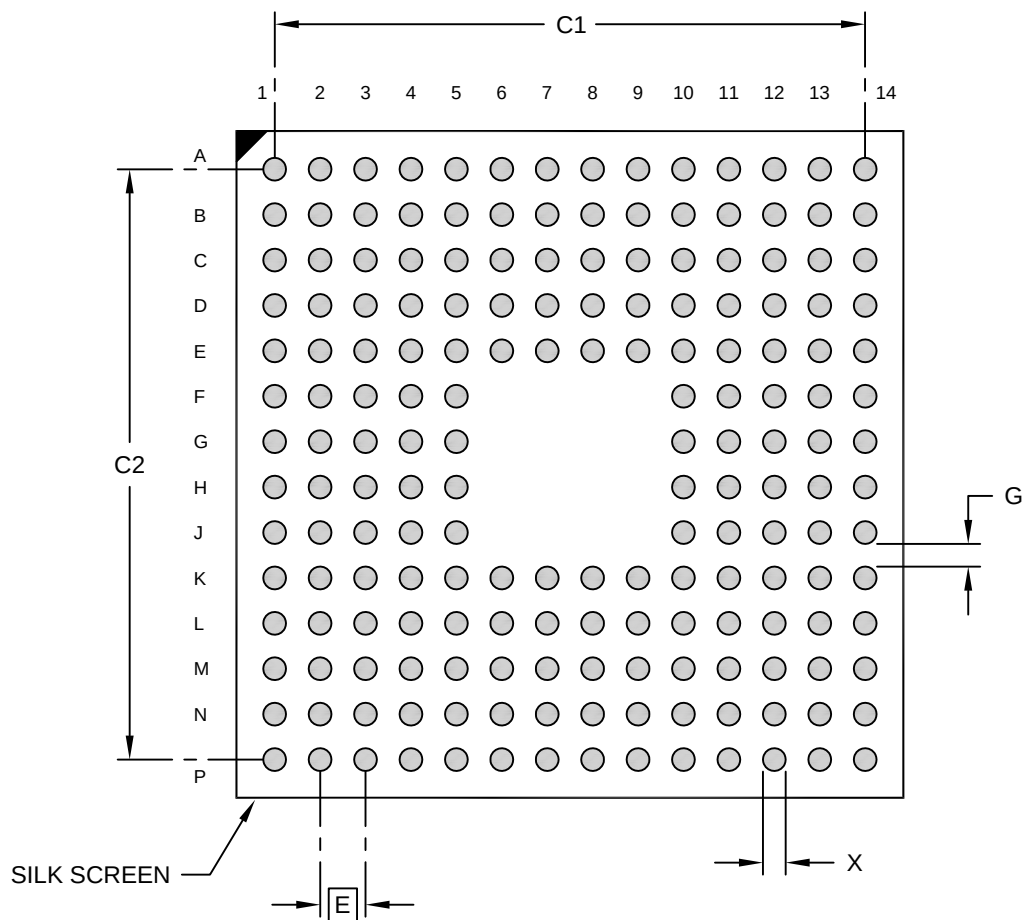
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	180		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.40
Ball Height	A1	0.25	-	0.35
Substrate Thickness	A2	0.26 REF		
Mold Cap Thickness	A3	0.70 REF		
Overall Length	D	12.00 BSC		
Overall Width	E	12.00 BSC		
Terminal Width	b	0.35	0.40	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

180-Ball Low Profile Fine Pitch Ball Grid Array (AGB) - 12x12x1.3mm Body [LFBGA] Atmel Legacy Global Package Code CDZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		10.40	
Contact Pad Spacing	C2		10.40	
Contact Pad Width (X160)	X			0.40
Contact Pad to Contact Pad	G	0.35		

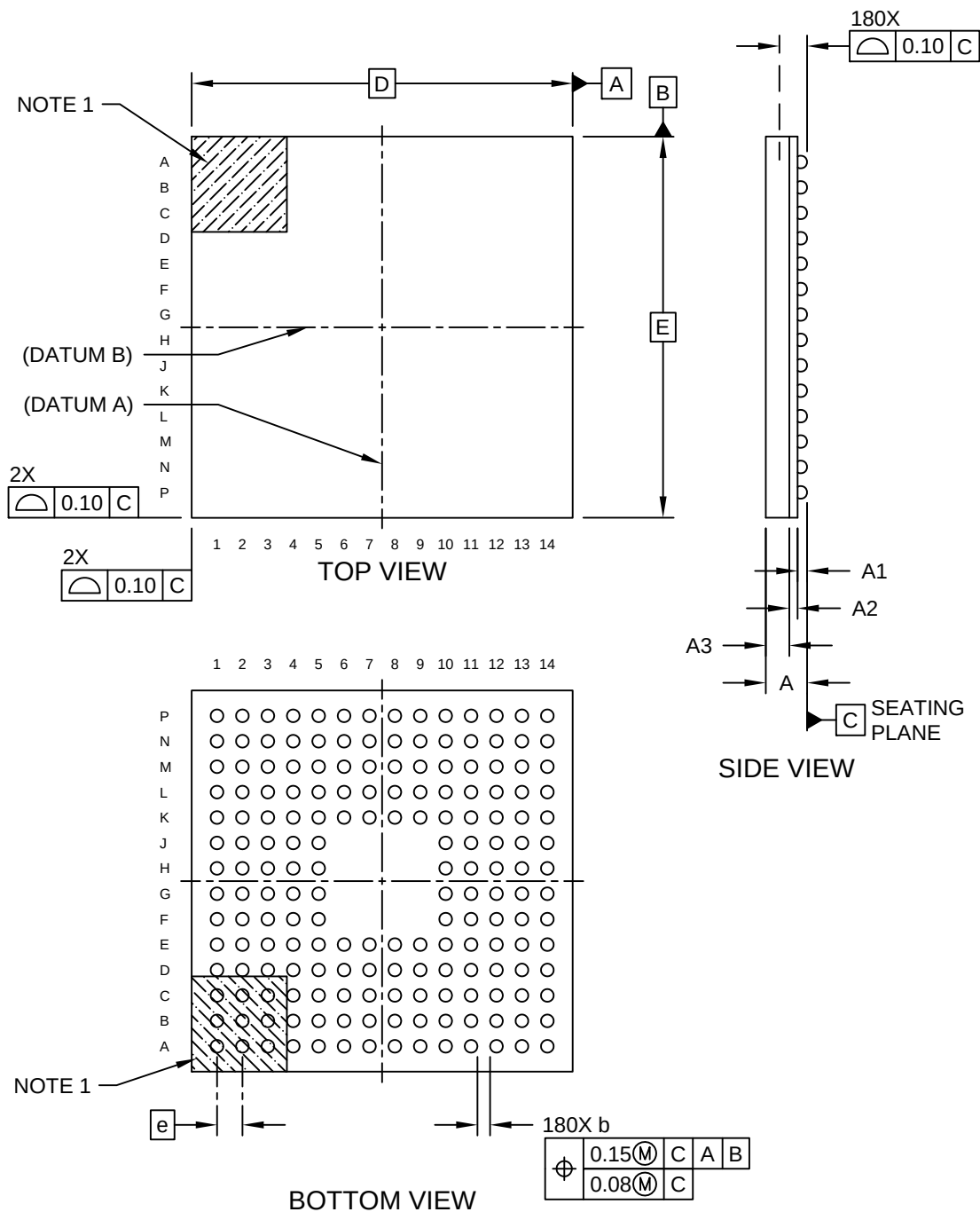
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23120-AGB Rev A

180-Ball Low Profile Fine Pitch Ball Grid Array (AHB) - 12x12x1.3mm Body [LFBGA] Atmel Legacy Global Package Code CDZ

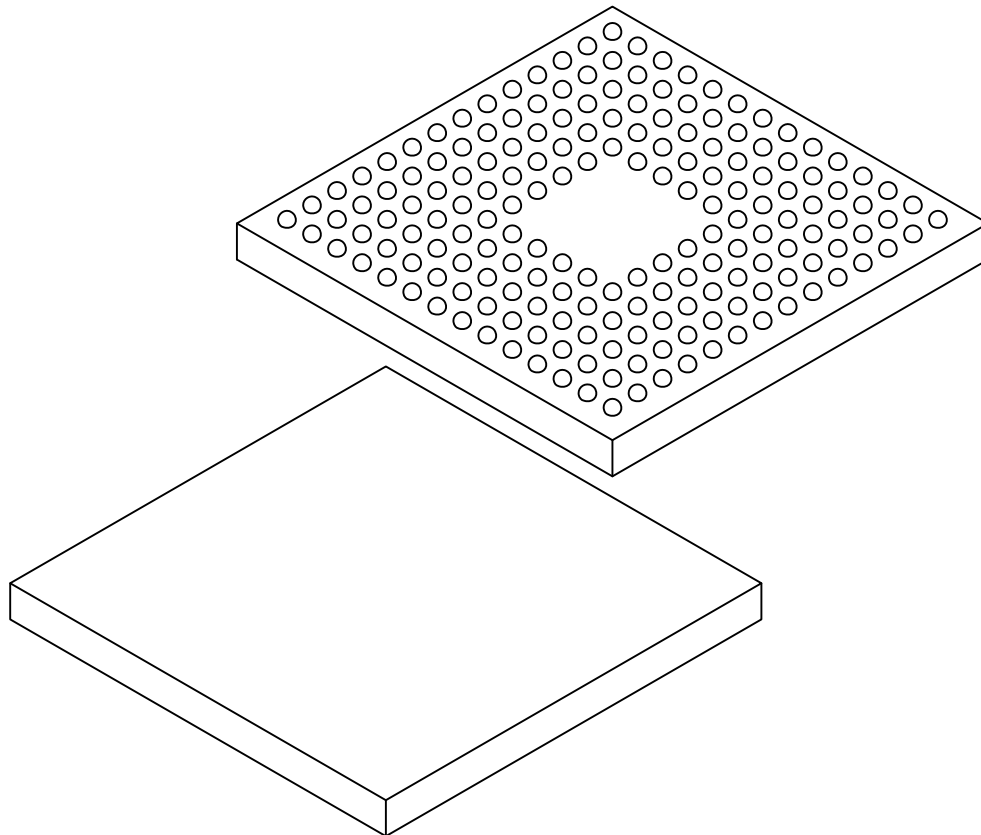
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-21120-AHB Rev A Sheet 1 of 2

180-Ball Low Profile Fine Pitch Ball Grid Array (AHB) - 12x12x1.3mm Body [LFBGA] Atmel Legacy Global Package Code CDZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



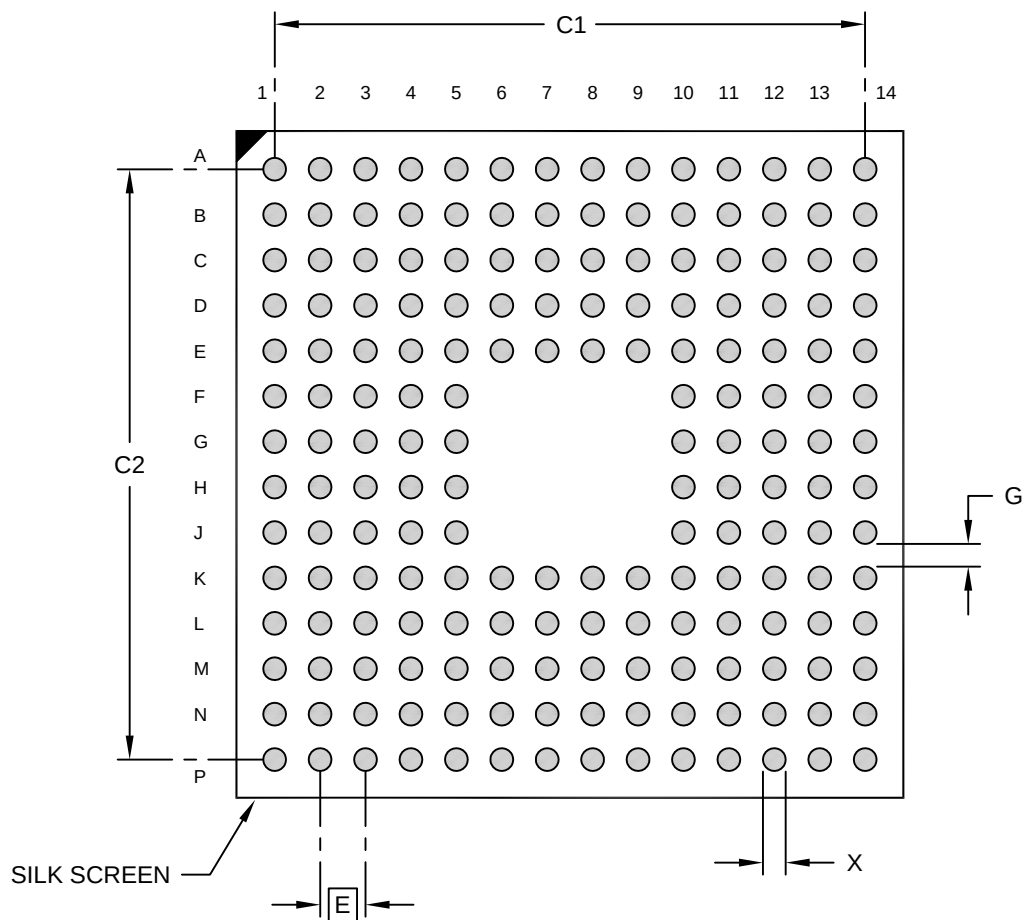
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	180		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.40
Ball Height	A1	0.25	-	0.35
Substrate Thickness	A2	0.26 REF		
Mold Cap Thickness	A3	0.70 REF		
Overall Length	D	12.00 BSC		
Overall Width	E	12.00 BSC		
Terminal Width	b	0.35	0.40	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

180-Ball Low Profile Fine Pitch Ball Grid Array (AHB) - 12x12x1.3mm Body [LFBGA] Atmel Legacy Global Package Code CDZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		10.40	
Contact Pad Spacing	C2		10.40	
Contact Pad Width (X160)	X			0.40
Contact Pad to Contact Pad	G	0.35		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23120-AHB Rev A